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Q1. Intelligent Customer Feedback Analysis

CODE

```
def preprocess(text):
    text = text.lower().strip()
    text = text.replace(".", "")
    text = text.replace(",", "")
    text = text.replace("!", "")
    text = text.replace("?", "")
    return text

def analyze(text):
    positive = ["good", "great", "excellent", "happy", "love", "nice", "best"]
    negative = ["bad", "poor", "worst", "hate", "sad", "terrible"]

    pos = 0
    neg = 0

    words = text.split()

    for word in words:
        if word in positive:
            pos += 1
        elif word in negative:
            neg += 1

    if pos > neg:
        result = "Positive"
    elif neg > pos:
        result = "Negative"
    else:
        result = "Neutral"

    return pos, neg, result

n = int(input())

for i in range(n):
    feedback = input()
    feedback = preprocess(feedback)
    p, q, sentiment = analyze(feedback)

    print("Positive Keywords:", p)
    print("Negative Keywords:", q)
    print("Sentiment:", sentiment)
```

INPUT

```
2
The product is Excellent, I love it!
The service is bad and worst.
```

OUTPUT

```
Positive Keywords: 2
Negative Keywords: 0
Sentiment: Positive
Positive Keywords: 0
Negative Keywords: 2
Sentiment: Negative
```

Q2. Comprehensive Password Strength Checker

CODE

```
def check_password(password):
    upper = 0
    lower = 0
    digit = 0
    special = 0

    for ch in password:
        if ch.isupper():
            upper += 1
        elif ch.islower():
            lower += 1
        elif ch.isdigit():
            digit += 1
        else:
            special += 1

    score = 0

    if len(password) ≥ 8:
        score += 1
    if upper > 0:
        score += 1
    if lower > 0:
        score += 1
    if digit > 0:
        score += 1
    if special > 0:
        score += 1

    if len(password) == 0:
        strength = "Weak"
    elif score ≤ 2:
        strength = "Weak"
    elif score == 3:
        strength = "Moderate"
    elif score == 4:
        strength = "Strong"
    else:
        strength = "Very Strong"

    print("Uppercase:", upper)
    print("Lowercase:", lower)
    print("Digits:", digit)
    print("Special Symbols:", special)
    print("Password Strength:", strength)

password = input().strip()
check_password(password)
```

INPUT

Abc@1234

OUTPUT

```
Uppercase: 1
Lowercase: 2
Digits: 4
Special Symbols: 1
Password Strength: Very Strong
```

Q3. Structured Server Log Error Analyser

CODE

```
def extract_errors(logs):
    errors = []
    for log in logs:
        if "error" in log.lower() or log.lower().startswith("error"):
            errors.append(log.strip())
    return errors

def remove_duplicates(errors):
    unique = []
    for error in errors:
        if error not in unique:
            unique.append(error)
    return unique

def summarize(errors):
    print("Total Errors:", len(errors))
    print("Unique Errors:")
    for error in errors:
        print(error)

n = int(input())

logs = []
for i in range(n):
    logs.append(input())

errors = extract_errors(logs)
unique_errors = remove_duplicates(errors)
summarize(unique_errors)
```

INPUT

```
5
INFO Server Started
ERROR Disk Full
WARNING Low Memory
ERROR Disk Full
ERROR Network Failure
```

OUTPUT

```
Total Errors: 2
Unique Errors:
ERROR Disk Full
ERROR Network Failure
```

Q4. Email Domain Frequency Analyser

CODE

```
n = int(input())

count = {}

for i in range(n):
    email = input().strip()

    if '@' in email:
        parts = email.split('@')
        if len(parts) == 2 and parts[1] != "":
            domain = parts[1].lower()

            if domain in count:
                count[domain] += 1
            else:
                count[domain] = 1

if len(count) == 0:
    print("No valid email domains found")
else:
    max_domain = ""
    max_count = 0

    for domain in count:
        print(domain, count[domain])
        if count[domain] > max_count:
            max_count = count[domain]
            max_domain = domain

    print(max_domain)
    print(max_count)
```

INPUT

```
5
alice@gmail.com
bob@yahoo.com
john@gmail.com
test@outlook.com
ram@gmail.com
```

OUTPUT

```
gmail.com 3
yahoo.com 1
outlook.com 1
gmail.com
3
```